

Outline

- 1 Getting a little “jumpy”: the basics of RDD
- 2 RDD: A Formal Examination
 - The data-generating process of RDD
 - Potential outcome perspective
 - Local experiment perspective
- 3 Estimation Issues for RDD
 - Predict conditional expectations
 - Select appropriate samples
- 4 RDD Validation
- 5 Other variants of RDD
- 6 Slides for R Notebook

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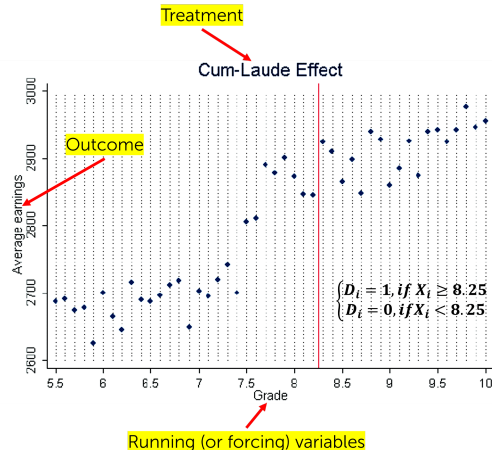
The cum laude effects on earnings at RSM

We cannot run an experiment!

How is the cum laude determined?

- Eligibility

$$D_i = \begin{cases} 1, & \text{if } X_i \geq 8.25 \\ 0, & \text{if } X_i < 8.25 \end{cases}$$



Exploiting variations from “rules”

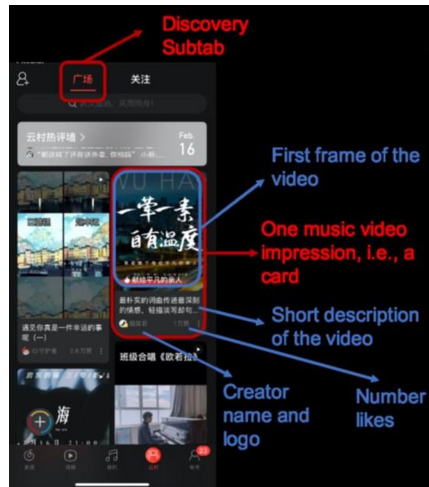
Policies, rules, or regulations create an assignment of treatment based on eligibility. . .

Examples of Eligibility Designs

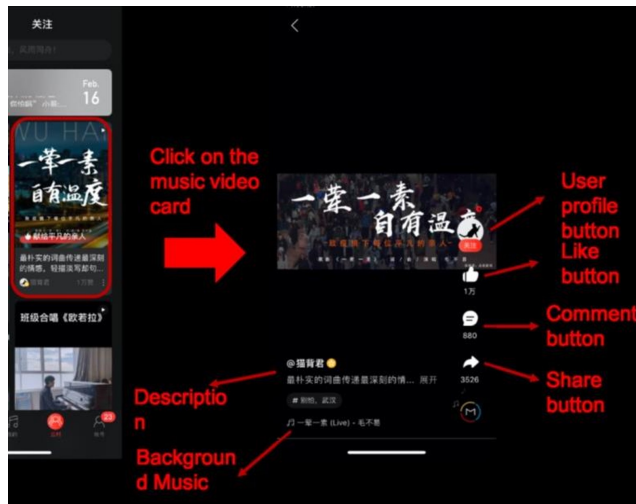
- Naturalization (years) on political integration of immigrants
- Loyalty program (consumer scores) on consumer purchases
- College admission (GPA scores) on future earnings
- Alcohol consumption (legal drinking age) on health outcomes

My own example - Netease Music

- Evaluating the effects of the recommender system at NetEase.
- NetEase has a streaming service.
- Recommend system called "discovery" in their app.



KPI:
click-through of
the
recommended
'cards'.



My own example - Netease Music

NetEase wanted to evaluate the performance of their recommender system.



Experiments

A random recommendation to the treatment group



RDD as a pre-experiment

Running variable: matching scores calculated from the recommender system

Treatment: Recommended on the discovery tab

Outcome: the clicks

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Some examples of “eligibility design”

Using the **product’s average numerical review score** as the running variable, **the platform’s rounding threshold** (at 4.25) as the cutoff to study the treatment of receiving a higher displayed star (e.g., 4.5 vs. 4.0) on the product’s subsequent sales.

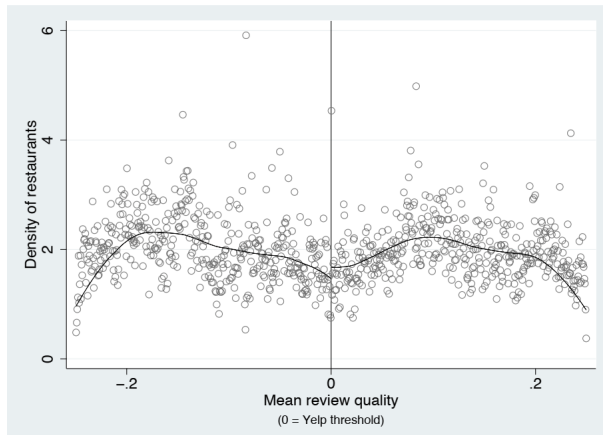
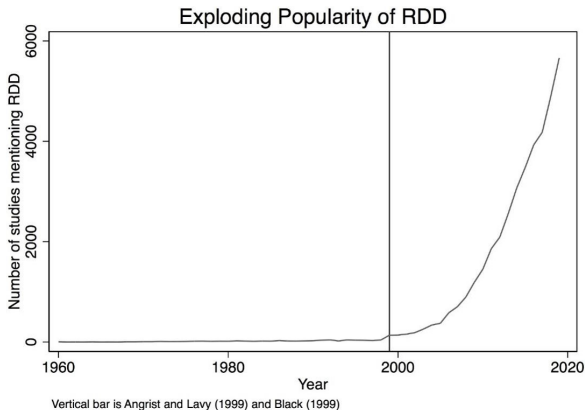


Figure 3: Anderson and Magruder (2012)

The origin of RDD and its explosion in recent years



The first research that used RDD is Thistlethwaite and Campbell (1960). The effect of merit awards on future academic outcomes with scores as the running variable.

Figure 4: Cunningham (2021)

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- 2 **RDD: A Formal Examination**
 - The data-generating process of RDD
 - Potential outcome perspective
 - Local experiment perspective
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- 4 RDD Validation
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① Getting a little “jumpy”: the basics of RDD

② RDD: A Formal Examination

The data-generating process of RDD

Potential outcome perspective

Local experiment perspective

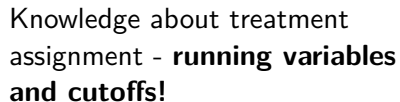
③ Estimation Issues for RDD

④ RDD Validation

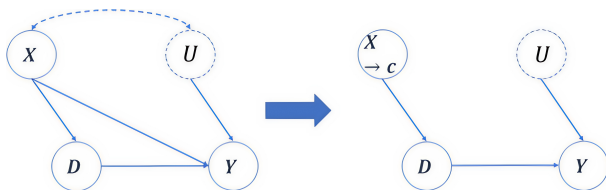
⑤ Other variants of RDD

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Regression Discontinuity Design



The DAG representation of RDD

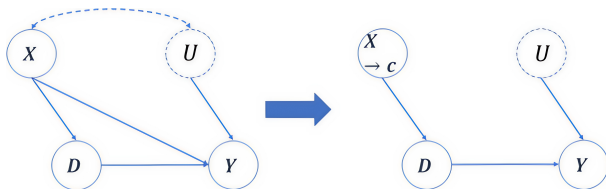


The treatment D is assigned based on the running variable X ($X \rightarrow D$).

X is likely to be associated with the confounders U ($X \leftrightarrow Z$).

For the DAG on the left, $P(Y | D)$ cannot be identified due to unblocked paths.

The DAG representation of RDD



The magic happens at the limit or the cutoff c .

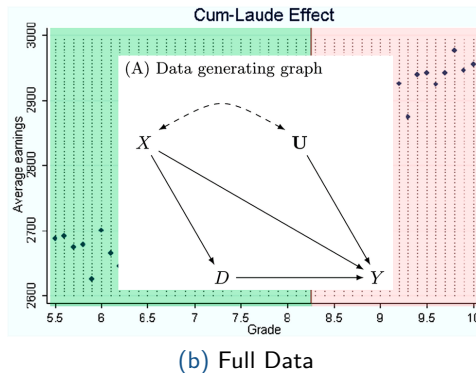
At the cutoff c , the treatment D only depends on c .

Therefore, X is excluded or the path $X \rightarrow Y$ is blocked by the cutoff c .

For the DAG on the right, $P(Y | D)$ is identified as the back-door paths between D and Y are blocked.

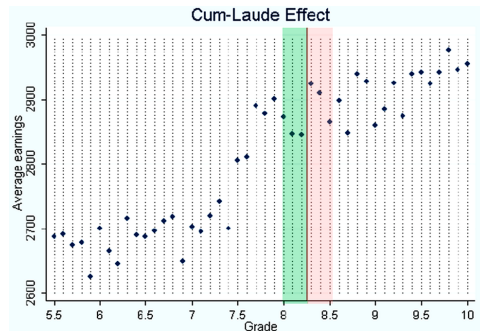
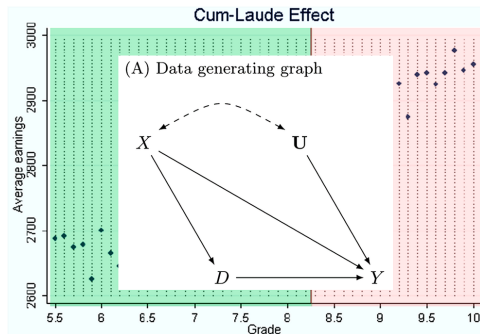
Data-generating process for RDD

Focus on a neighborhood close to the cutoff point.



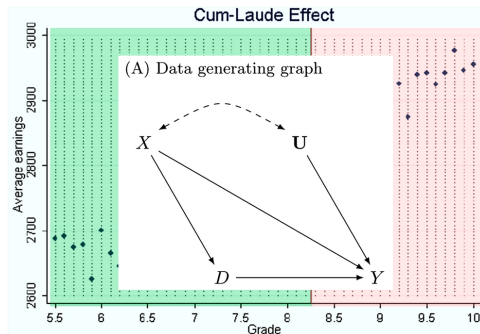
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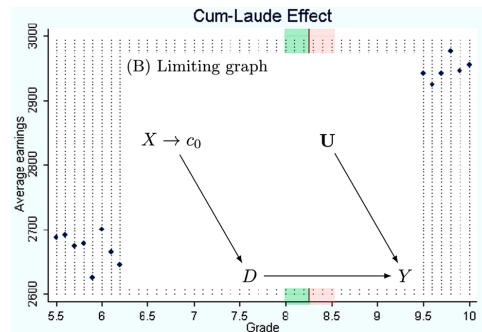


Data-generating process for RDD

Focus on a neighborhood close to the cutoff point.



(e) Full Data



(f) RDD Data

① Getting a little “jumpy”: the basics of RDD

② RDD: A Formal Examination

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Potential outcome perspective

Local experiment perspective

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Identification with potential outcomes perspective

Suppose we have a running variable X_i with the cutoff c , the treatment D_i and the outcome Y_i .

- Given the setup of RDD, the treatment assignment is:

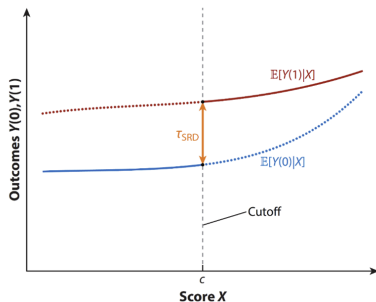
$$P(D_i = 1 \mid X_i \geq c) = 1$$

$$P(D_i = 1 \mid X_i < c) = 0$$

- The potential outcome and the observed outcome are:

$$Y_i = Y_i^1 D_i + Y_i^0 (1 - D_i)$$

Identification with potential outcomes perspective



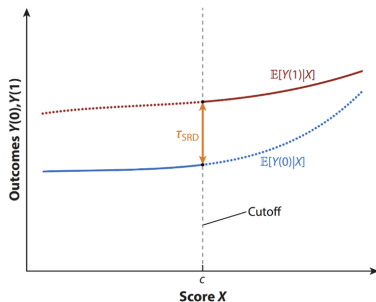
The treatment effect is identified at the cutoff

$$\tau_{RDD} = E(Y_i^1 - Y_i^0 \mid X_i = c)$$

To define τ_{RDD} in terms of limits:

$$\tau_{RDD} = \lim_{X_i \downarrow c} E(Y_i^1 \mid X_i = x) - \lim_{X_i \uparrow c} E(Y_i^0 \mid X_i = x)$$

Identification with potential outcomes perspective



$$\tau_{RDD} = \lim_{X_i \downarrow c} E(Y_i^1 | X_i = x) - \lim_{X_i \uparrow c} E(Y_i^0 | X_i = x)$$

For τ_{RDD} to exist, the two limits must exist.

A sufficient condition: **both $E(Y_i^1 | X_i = x)$ and $E(Y_i^0 | X_i = x)$ are continuous.**

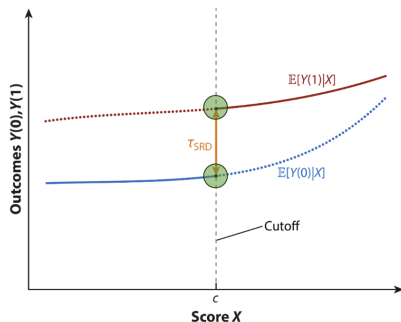
Understanding the continuity assumption

So, the overlap assumption is violated. **What are the consequences?**

$$\tau_{RDD} = \underbrace{\lim_{X_i \downarrow c} E(Y_i^1 | X_i = x)}_{P(D_i=1|X_i > c)=1} - \underbrace{\lim_{X_i \uparrow c} E(Y_i^0 | X_i = x)}_{P(D_i=1|X_i \leq c)=0}$$

- Only $\lim_{X_i \downarrow c} E(Y_i^1 | X_i = x)$ is observed, as $P(D_i = 1 | X_i > c) = 1$.
- Extra assumptions are needed!

Understanding the continuity assumption



We need to somehow “infer”
 $\lim_{X_i \uparrow c} E(Y_i^0 | X_i = x).$

Learn this function

$$E(Y_i^0 | X_i < c) = \sum_{X_i < c} Y_i^0 P(Y_i^0 | X_i).$$

Extrapolate to $E(Y_i^0 | X_i = c)$ under the **continuity assumption**.

Estimation procedure with continuity assumption

Algorithm 1: RD estimation with continuity assumption

Input: Running variable X_i , Treatment variable D_i , Cutoff c , Outcome Y_i

Output: Estimated treatment effect τ_{RDD}

- 1 Learn $\mathbb{E}(Y_i^0 \mid X_i < c)$ using observations with $X_i < c$;
- 2 Learn $\mathbb{E}(Y_i^1 \mid X_i \geq c)$ using observations with $X_i \geq c$;
- 3 Compute $\mathbb{E}(Y_i^0 \mid X_i = c)$ using the learned function from Step 1;
- 4 Compute $\mathbb{E}(Y_i^1 \mid X_i = c)$ using the learned function from Step 2;
- 5 Compute treatment effect:

$$\tau_{RDD} = \mathbb{E}(Y_i^1 \mid X_i = c) - \mathbb{E}(Y_i^0 \mid X_i = c)$$

① Getting a little “jumpy”: the basics of RDD

② RDD: A Formal Examination

The data-generating process of RDD

Potential outcome perspective

Local experiment perspective

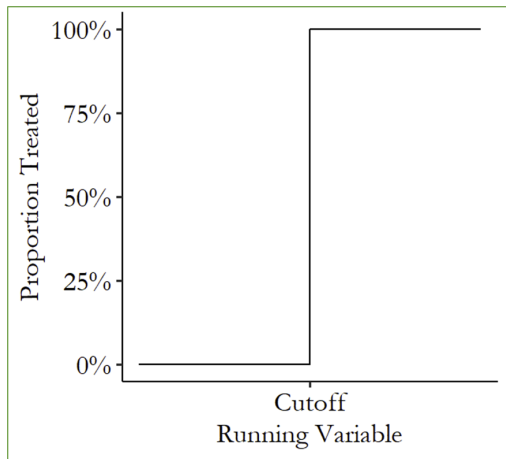
③ Estimation Issues for RDD

④ RDD Validation

⑤ Other variants of RDD

⑥ Slides for R Notebook

The CRE and RDD connection



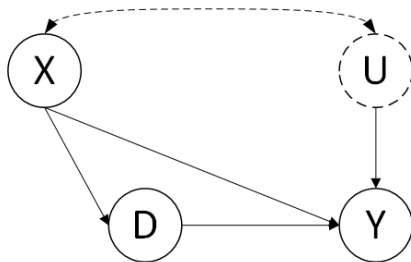
A CRE is a special case of RDD:

- Running variable: $X \sim U(0, 1)$.
- Cutoff: $c = 0.5$.
- Treatment assignment:

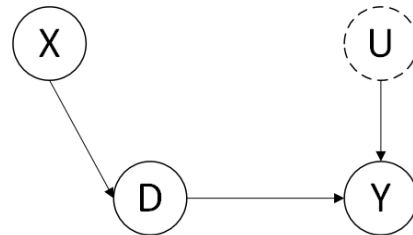
$$D = \begin{cases} 1 & \text{if } X \geq c \\ 0 & \text{if } X < c \end{cases}$$

The CRE and RDD connection

A CRE is a special case of RDD.



(g) DAG for RDD



(h) DAG for CRE

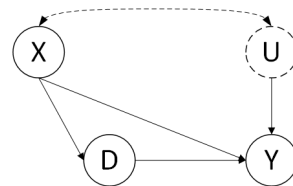
The CRE and RDD connection

What's the difference between RDD and CRE?

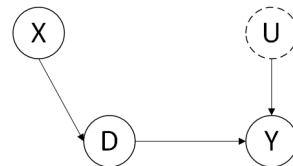
- Consider the equation system:

$$\begin{cases} Y = \tau D + \delta_1 Z + U \\ D = 1(X \geq c) \\ X = \delta_2 Z + V \end{cases}$$

- For CRE, $\delta_2 = 0$ (exclusion) and $V \perp U$ (exogeneity).

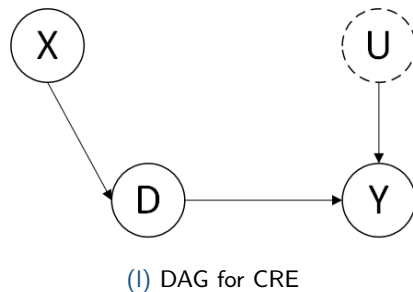
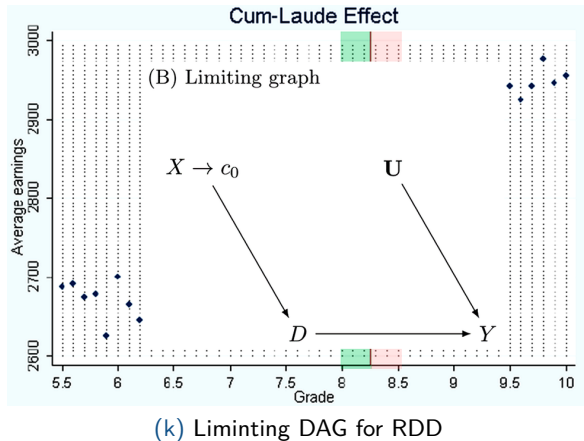


(i) DAG for RDD



(j) DAG for CRE

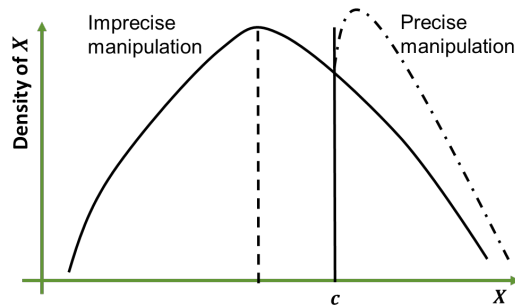
RDD as a local randomized experiment



RDD as a local randomized experiment

What assumption is needed for RDD as a local experiment?

- The local treatment D should be “as good as random”.
- **No manipulation:** people do not have precise control over the running variable.



RDD Estimation with no manipulation assumption

Algorithm 2: Local Experiment RDD Estimation

Input: Running variable X_i , Treatment variable D_i , Cutoff c , Outcome Y_i

Output: Estimated treatment effect $\tau_{RDD} = \mathbb{E}(Y_i^1 \mid X_i = c) - \mathbb{E}(Y_i^0 \mid X_i = c)$

- 1 Select window $(c - h_0, c + h_1)$, where $h_0, h_1 > 0$;
- 2 Trim data to include only observations with $X_i \in (c - h_0, c + h_1)$;
- 3 Estimate treatment effect with difference-in-means:

$$\hat{\tau}_{RDD} = \frac{1}{n_T} \sum_{i \in T} Y_i - \frac{1}{n_C} \sum_{i \in C} Y_i$$

where $T = \{i \mid X_i \in (c, c + h_1)\}$ and $C = \{i \mid X_i \in (c - h_0, c)\}$.

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Predict conditional expectations

Select appropriate samples

4 RDD Validation

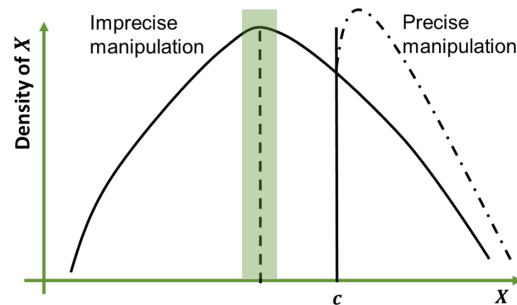
5 Other variants of RDD

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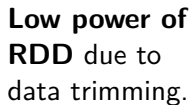
Which estimation procedure?

In practice, the procedure with potential outcomes is commonly used.

Why? **Low power of RDD** due to data trimming - a close neighborhood around the cutoff.



Power Comparison: RDD vs CRE



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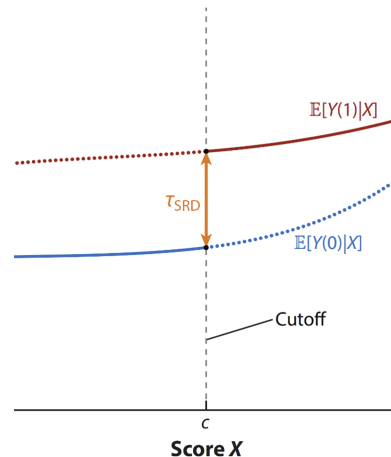
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How to predict conditional expectations?

Under the continuity assumption, we can estimate the conditional expectation function.

- Using data above the cutoff ($X \geq c$) to estimate $E(Y^1|X)$.
- Using data below the cutoff ($X < c$) to estimate $E(Y^0|X)$.



Parametric models for predicting expectations

Parametric models for $E(Y^1|X)$ and $E(Y^0|X)$.

- Linear:

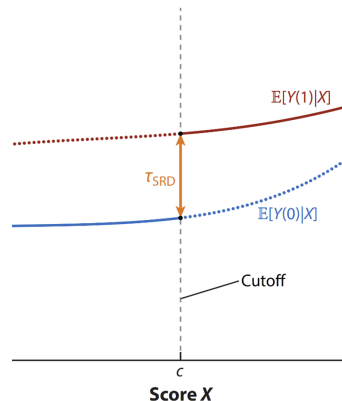
$$Y_i = \alpha + \beta D_i + \gamma X_i + e_i.$$

- With interaction:

$$Y_i = \alpha + \beta D_i + \gamma X_i + \theta D_i X_i + e_i.$$

- Polynomial:

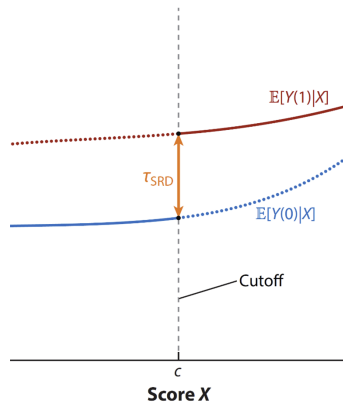
$$Y_i = \alpha + \beta D_i + \gamma_1 X_i + \gamma_2 X_i^2 + \theta_1 D_i X_i + \theta_2 D_i X_i^2 + e_i.$$



A side note about the order of the polynomials

Gelman and Imbens (2019) pointed out several problems and recommend using only polynomials up to the second degree (quadratic).

- Polynomial estimation can be noisy with polynomials of higher order.
- Estimates can be sensitive to the degree of the polynomial.
- Possible bias from higher-order polynomials - Confidence intervals with bad coverage.

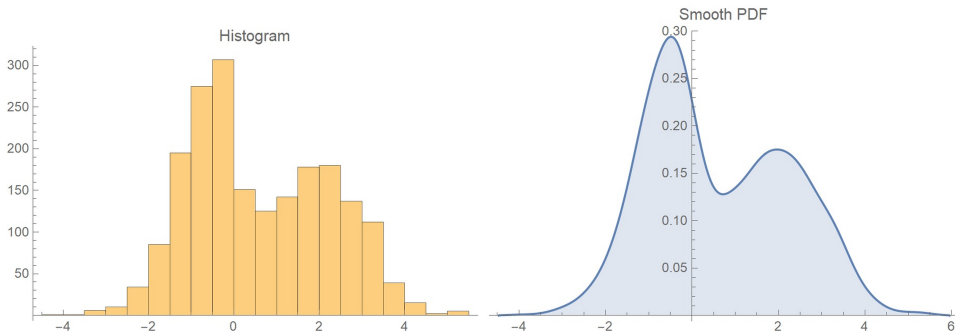


Regression Discontinuity Design

-
- A histogram showing the density of the number of children. The x-axis is labeled 'X' and ranges from approximately -4 to 4, with major ticks at -3, 0, and 3. The y-axis is labeled 'Density' and ranges from 0.00 to 0.20, with major ticks at 0.00, 0.05, 0.10, 0.15, and 0.20. The histogram bars are light gray with black outlines. A solid blue curve is overlaid on the histogram, representing a fitted normal distribution. The curve is bell-shaped and centered around 0, with its peak density slightly above 0.18.

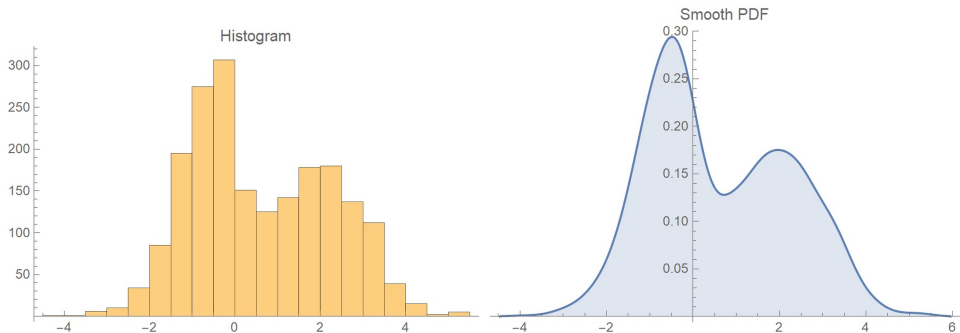
A glimpse at non-parametric estimation

- In simple cases, it's straight forward, e.g., estimate the mean $\hat{E}(X) = N^{-1} \sum X_i$ (no parameters).
- For discrete variables, the probability density function is just $p_1(X) = N_1^{-1} \sum D(X_i \in C_1)$.



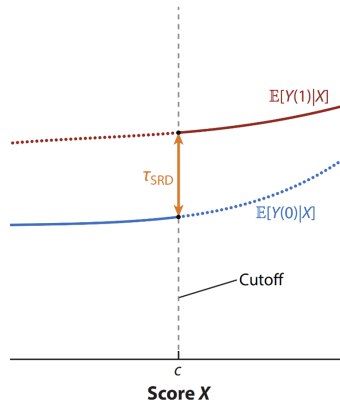
A glimpse at non-parametric estimation

- For continuous variable, we use **binning** as in producing a histogram.
- To further use small **kernels** (a smoothing function) to smooth out the histogram.



Summary: predicting expectation functions

- The true functional form is unknown - **no ground truth**.
- **No guarantee** that a non-parametric model is better, or a complex model is better.
- In practice, we often try **different models** to show the robustness.
- However, if the “**jump**” is **big**, the functional forms probably do not matter.



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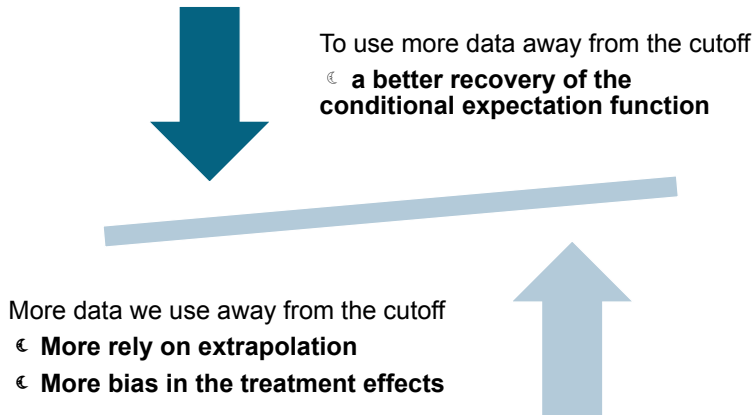
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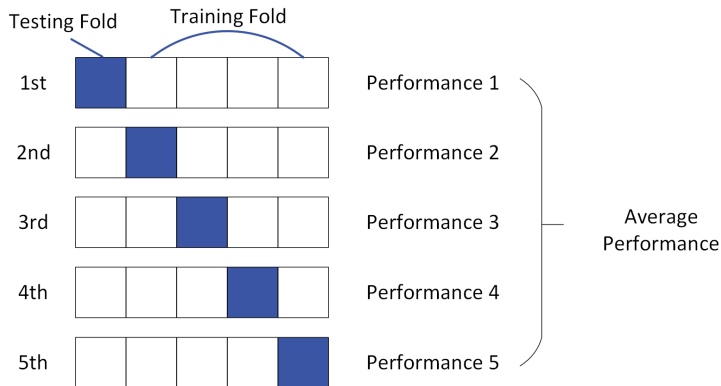
④ RDD Validation

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Balancing in-sample vs. out-sample fitting.



The cross-validation for RDD estimation

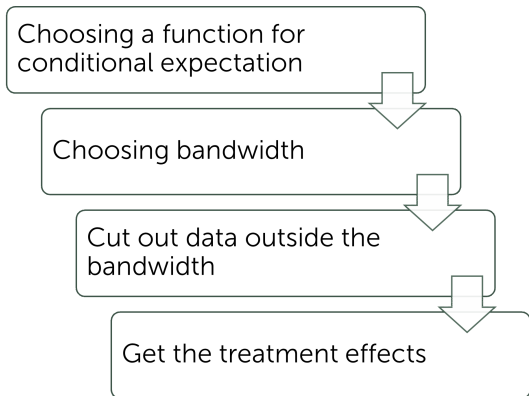
Algorithm 3: RDD Bandwidth Selection via Cross-Validation

Input: Grid of bandwidth parameters $\{\rho_j\}_{j=1}^M$, Number of data folds K

Output: Optimal bandwidth ρ^*

- 1 Randomly split the full data into K folds;
 - 2 **foreach** ρ_j in $\{\rho_1, \rho_2, \dots, \rho_M\}$ **do**
 - 3 Set bandwidth $h_j = \rho_j \cdot SD(X_i)$;
 - 4 **for** $k = 1$ **to** K **do**
 - 5 Trim both training and validation sets with where $|X_i - c| \leq h_j$;
 - 6 Fit $E[Y_i | X_i, D_i]$ using training sets $K \setminus \{k\}$;
 - 7 Compute performances (e.g., MSE) on validation set k ;
 - 8 Compute average performance over K folds for h_j ;
 - 9 Select ρ^* with the best performance;
-

Summary: the general estimation procedure for RDD



Placeholder for RDD GIF.

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Examining the continuity assumption

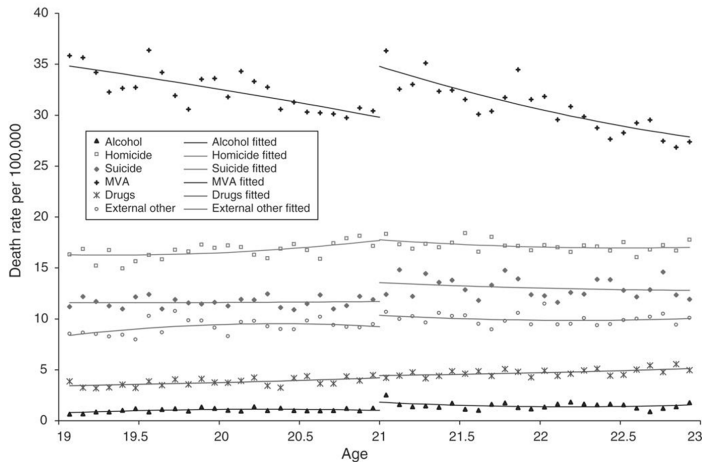


Figure 5: Captenter and Dobkin (2009)

We do not observe the counterfactual outcomes.

Use outcomes that should be / not be affected by the treatment as “consistency tests.”^a

^aMore on placebo / consistency tests in the last session.

Examining the continuity assumption

Why don't we examine the distribution of the running variable X_i directly?

Expect no discontinuity around the cutoff (“no manipulation”) - McCray density test.

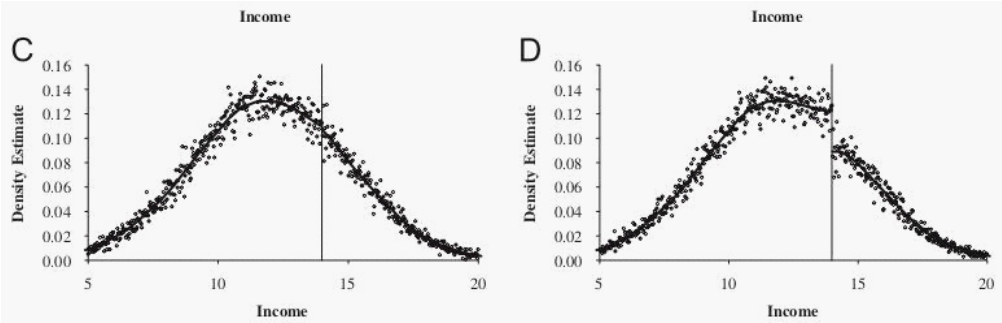
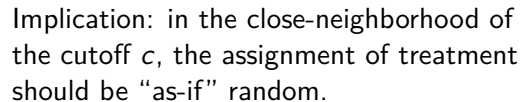


Figure 6: McCray (2008)

Regression Discontinuity Design



Balance tests for RDD

If we have local randomization, the characteristics of people should be **balanced above and below the cutoff**.

Balance test: covariates are balanced around the cutoff.

Variable	Std. Mean Difference	P-value
<i>Age</i>		
25 - 44	0.018	0.849
45 - 64	-0.018	0.849
65 $\geq$	-0.087	0.360
<i>Education Level</i>		
Primary	0.078	0.409
Secondary	0.058	0.542
Post-secondary	-0.151	0.111
<i>Household Wealth Index</i>		
Q1	-0.056	0.538
Q2	0.00	1.000
Q4	0.142	0.135
Q5	-0.154	0.104
<i>Region</i>		
Northern	-0.155	0.103
Middle	0.158	0.102
<i>Other</i>		
Household size	-0.007	0.942
Household size squared	-0.051	0.599
Overweight(25 $\leq$ BMI $\leq$ 29.9)	0.00	1.000
Renewed insurance at least once	0.111	0.241
Female	0.036	0.701
Married	0.051	0.593
Urban	0.032	0.737
At least one child in the household	0.00	1.000

Important: how to interpret τ_{RDD} ?



In RDD, treatment effects are local average treatment effects or LATE.



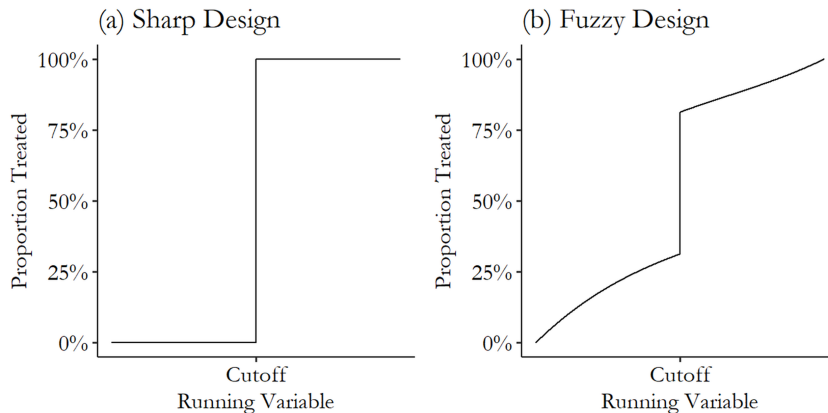
Not average treatment effect, but rather the effect of getting the treatment for units that were close to the cutoff.



This is the price we pay for a non-random assignment.

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Fuzzy RDD

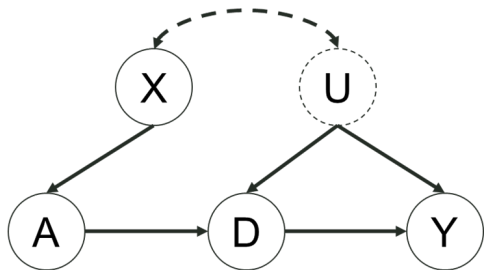


Fuzzy RDD: the treatment has **non-compliance**.

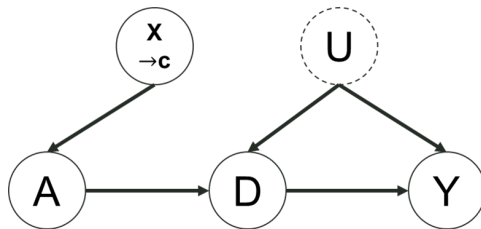
Figure 7: Comparing Sharp vs. Fuzzy RDD

DAGs for fuzzy RDD

- The assignment A depends on the cutoff of the running variable.
- The actual treatment D depends on the assignment.



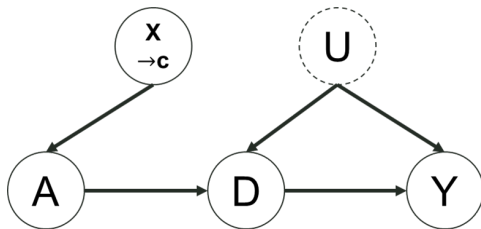
(a) Data generating DAG



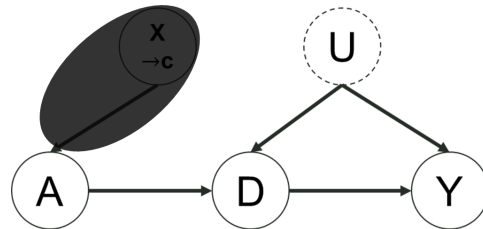
(b) Limiting DAG

DAGs for fuzzy RDD

- In the limit $X \rightarrow c$, the DAG becomes just like in the non-compliance setting.
- Assignment A becomes an instrument for the treatment D .



(c) Limiting DAG



(d) IV estimation

Regression kinked design (RKD)

- RDD centered around a **discontinuous jump** in the treatment (binary).
- Many policy tools have shifts in **the treatment intensity** based on the running variables, rather than jumps.



Income taxes



Usage-based pricing



Tiered loyalty program

Figure 8: Examples of RKD

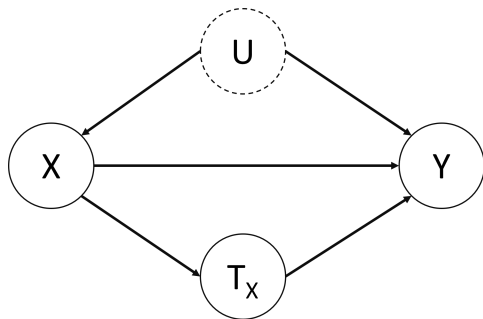
Regression Discontinuity Design

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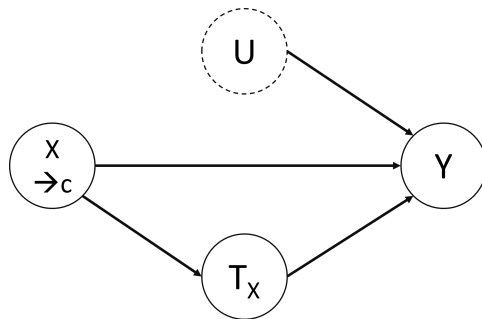
Identification for RKD

The conditional expectations of the potential outcomes $E(Y|X)$ is **continuously differentiable** at the cutoff c .

Implications: **Only the treatment rule leads to the change in slopes.**



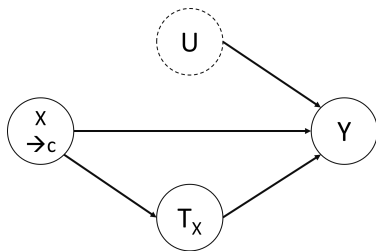
(a) Data generating DAG



(b) Limiting DAG

Identification results of RKD

The local marginal effect of treatment intensity $T(X)$ on outcome Y at the cutoff c .



$$\tau_{RKD} = \frac{\Delta \left(\frac{\partial \mathbb{E}[Y|X]}{\partial X} \right)}{\Delta \left(\frac{\partial T(X)}{\partial X} \right)} \Bigg|_{X=c}$$

In detail:

$$\tau_{RKD} = \frac{\lim_{x \downarrow c} \frac{\partial}{\partial x} \mathbb{E}[Y | X = x] - \lim_{x \uparrow c} \frac{\partial}{\partial x} \mathbb{E}[Y | X = x]}{\lim_{x \downarrow c} \frac{\partial}{\partial x} T(X) - \lim_{x \uparrow c} \frac{\partial}{\partial x} T(X)}$$

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Keywords auction on Google

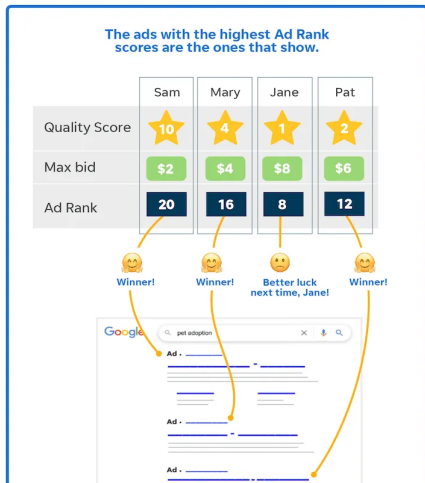


Objective: **to measure the value of positions.**

Observations:

- The position is determined by AdRank.
- For two positions j and $j + 1$, we must have $\text{AdRank}_j - \text{AdRank}_{j+1} \geq 0$.

Using RDD to evaluate position effects



- **Running variable:**
 $\Delta \text{AdRank}_{ij} = \text{AdRank}_i - \text{AdRank}_j$.
- **Cutoff:** if $\Delta \text{AdRank}_{ij} > 0$.
- **Treatment:** Rank at position $k + 1$ vs. k .
- **Outcome:** Revenue or other variables.

Regression Discontinuity Design

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